

Local Curvature, Trace Periods, and Galois Covers of Black-Hole Horizons

Valuations, Landen–Trace reductions, and the
degree-20 entropy crown

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Abstract

Multi-horizon black-hole thermodynamics is organised here at three successive scales. Locally, near normal-crossing collapses of thermodynamic coordinates, diagonal inverse-channel metrics admit an exact directional decomposition of scalar curvature. A master quadratic form on the metric valuation matrix controls normal-vertex coefficients and yields the generic order-three and parity-fixed order-four face laws, proper-distance rates, and a complete cancellation hierarchy in arbitrary codimension. At the period scale, the complementary complete integrals that arise from those channels reduce by explicit degree-two isogenies to a common elliptic curve E_{128} : each differential splits into a non-torsion third-kind trace form Θ and an elementary logarithmic anti-translation term, with exact primary and dual anti-periods. Globally, the horizon entropy pair is a Galois cover of the observables. For Kerr–Newman the cover is quadratic and the left/right dictionary is its $\mathbb{Z}/2$ grading; at four charges the seed-mass fiber is an S_5 quintic, so Abel–Ruffini obstructs radical expressions for mass and entropy while the area product remains rational. Above the quintic the static ordered-horizon field has degree 20 and normal closure C_2^2 wr S_5 ; rotation lifts to C_2^3 wr S_5 . The paper is written as a single sequential argument from local valuations through period reduction to Galois descent.

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1 Introduction

The thermodynamics of multi-horizon black holes is usually presented as a collection of exact identities: sector entropies and temperatures, first laws on both horizons, negative inner temperatures, universal area products, and hidden-conformal data [1, 2, 3, 4, 5, 6, 7, 8, 9]. Those identities are almost always derived by writing the outer and inner entropies $S_{\pm}$ explicitly in an auxiliary solution-generating frame—a seed mass m and boost parameters δ_i [11, 12]—and then forming the combinations of interest. The present paper starts from a different organising question:

Which objects of the multi-horizon story are intrinsic to the physical observables, and which exist only after one adjoins roots, channels, or auxiliary frames?

The answer is developed at three successive scales, each preparing the next.

Local scale. Thermodynamic coordinates on the Kerr–Newman family collapse along normal-crossing divisors: vanishing temperature, angular velocity, or electric potential. Near those strata the metric takes an inverse-channel form, and the scalar curvature develops controlled poles. Section 2 proves that for diagonal divisor-monomial metrics the scalar curvature decomposes into finitely many directional channels. The leading normal coefficient is a master quadratic form on one column of the metric valuation matrix. From that single identity one obtains the generic order-three law, the parity-fixed order-four law with coefficient $-m(m+5)/B$, proper-distance normalisations, Newton polyhedra in arbitrary codimension, and an exhaustive cancellation hierarchy. No global curvature expression is ever required.

Period scale. Directional channels do not stop at pole orders. Complementary complete integrals built from those channels define a pair of meromorphic differentials $\omega_{\pm}$ on genus-one curves $C_{\pm}$. Section 3 reduces both differentials by explicit degree-two isogenies onto one common target curve E_{128} of j -invariant 128. On that curve the even combination descends to a non-torsion third-kind differential Θ , while the odd combination closes as an elementary logarithmic form. Exact anti-periods fix the elementary corrections: -4π in the primary channel and a vanishing Cauchy principal value in the dual channel. The two channels share the same target and the same pole, but travel different relative paths—a comparison left open, and cleanly separated from what is proved.

Global scale. The same even/odd pattern reappears algebraically for the entropy pair itself. Section 4 treats (S_+, S_-) as a Galois cover of the asymptotic charges. For Kerr–Newman the cover is the rational quadratic

$$S^2 - e_1 S + e_2 = 0, \quad e_1 = 2\pi(2M^2 - Q^2), \quad e_2 = \pi^2(4J^2 + Q^4), \quad (1)$$

and the classical left/right dictionary is the $\mathbb{Z}/2$ grading of that cover: invariants descend to the observables, odd data does not. At four charges the area product remains rational, but the seed-mass fiber is an irreducible S_5 quintic. Abel–Ruffini therefore obstructs radical expressions for the mass and the individual entropies; universality of the product is Galois descent. Above the quintic, Kummer data produce a degree-20 ordered-horizon field with maximal wreath normal closures, and inertia is classified by coloured cycle data on the incidence cover.

Reading order. The sections are meant to be read in order. Local valuations identify the geometric channels; period reduction separates what descends from what remains elementary or path-dependent; the Galois theory of the entropy cover is the same separation executed on the thermodynamic observables. Notation is kept consistent across scales: Θ always denotes the descended trace differential on E_{128} ; the Galois-odd temperature-entropy covariant is written $\vartheta := ST$; local transverse first drift is written $\mathcal{D}$. Conventions are $G = \hbar = c = 1$, $S = A/4$, and scalar curvature normalised so that the unit two-sphere has $R = +2$. The fundamental Kerr–Newman relation is

$$M^2 = U(S, J, Q) = \frac{S}{4\pi} + \frac{\pi J^2}{S} + \frac{Q^2}{2} + \frac{\pi Q^4}{4S}, \quad (2)$$

with potentials $T = U_S/(2M)$, $\Omega = U_J/(2M)$, $\Phi = U_Q/(2M)$.

Relation to prior work. Sector laws, Smarr formulae, negative inner temperatures, and area products are classical [1, 2, 3, 4, 5, 6]; hidden-conformal data appear in [7, 8, 9]. The contribution here is the derivation principle and the three-scale spine: local valuations, period reduction, and non-abelian obstruction for the non-extremal cover. Extremal arithmetic (attractors, class numbers, CM) [14, 15] is complementary. Novelty claims remain provisional pending a full prior-art sweep.

2 Local curvature valuations

The first scale is purely local. Thermodynamic faces of the Kerr–Newman family— $T = 0$, $\Omega = 0$, $\Phi = 0$ —are normal-crossing divisors on which the metric collapses in a characteristic inverse-channel pattern. This section develops the curvature calculus needed to read those faces without assembling a global scalar-curvature formula.

2.1 Diagonal divisor-monomial metrics

Definition 2.1 (Diagonal divisor-monomial metric germ). On the positive orthant $z_a > 0$ with coordinates $(t^1, \dots, t^n) = (z_1, \dots, z_k, y_{k+1}, \dots, y_n)$ and divisor $D = \bigcup_{a=1}^k \{z_a = 0\}$, a *diagonal divisor-monomial metric germ* is

$$g = \sum_{i=1}^n g_i (dt^i)^2, \quad g_i = h_i(z, y) z^{P_i}, \quad z^{P_i} := \prod_{a=1}^k z_a^{p_{ia}},$$

where each unit h_i extends smoothly and positively to the corner. The matrix $P = (p_{ia}) \in M_{n \times k}(\mathbb{R})$ is the *metric valuation matrix*.

Definition 2.2 (Curvature order). Write $\text{ord}_{D_a}(f) = \lambda$ when $f = z_a^\lambda u$ with $u|_{D_a} \neq 0$, and set $\kappa_{D_a}(f) := -\text{ord}_{D_a}(f)$. Thus $\kappa > 0$ is a pole. For a positive weight w , $\text{ord}_w(z^\alpha) = \langle w, \alpha \rangle$ and $\kappa_w = -\text{ord}_w$.

Definition 2.3 (Parity-fixed inverse-channel face). A face $x = 0$ is *parity-fixed* when every metric component is even in an adapted chart. The inverse-channel pattern in dimension $m + 1$ is

$$g_x = Bx^2 + O(x^4), \quad g_{y_\alpha} = A_\alpha x^{-2} + O(1), \quad \alpha = 1, \dots, m,$$

with $B, A_\alpha > 0$.

These are exactly the local models needed later for the Kerr–Newman strata $T = 0$ (generic quadratic collapse) and $\Omega = 0$, $\Phi = 0$ (parity-fixed inverse channels).

2.2 Directional curvature channels

For a diagonal metric set $H_i = \sqrt{g_i}$, $u_i = \log H_i = \frac{1}{2} \log g_i$, and $\beta_{ij} = (\partial_i H_j)/H_i$ for $i \neq j$.

Lemma 2.4 (Lamé formula).

$$R = -2 \sum_{i < j} \frac{1}{H_i H_j} \left(\partial_i \beta_{ij} + \partial_j \beta_{ji} + \sum_{\ell \neq i, j} \beta_{\ell i} \beta_{\ell j} \right).$$

Proof. The nonzero Christoffel symbols of an orthogonal metric are

$$\Gamma_{ii}^i = \partial_i \log H_i, \quad \Gamma_{ij}^i = \partial_j \log H_i, \quad \Gamma_{jj}^i = -\frac{H_j \partial_i H_j}{H_i^2} \quad (i \neq j),$$

and the symbols with three distinct indices vanish. Substitution into R_{jij}^i gives the coordinate-plane sectional curvature

$$K_{ij} = -\frac{1}{H_i H_j} \left[\partial_i \beta_{ij} + \partial_j \beta_{ji} + \sum_{\ell \neq i, j} \beta_{\ell i} \beta_{\ell j} \right].$$

Since $R = 2 \sum_{i < j} K_{ij}$, the claim follows. $\square$

Lemma 2.5 (Directional channels). *Define*

$$\mathcal{Q}_j(u) := \sum_{i \neq j} \left[\partial_j^2 u_i + (\partial_j u_i)^2 - (\partial_j u_j)(\partial_j u_i) \right] + \sum_{\substack{r < s \\ r, s \neq j}} (\partial_j u_r)(\partial_j u_s). \quad (3)$$

Then

$$R[g] = -2 \sum_{j=1}^n e^{-2u_j} \mathcal{Q}_j(u) = -2 \sum_{j=1}^n \frac{\mathcal{Q}_j(u)}{g_j}. \quad (4)$$

Proof. Because $\beta_{ij} = e^{u_j - u_i} \partial_i u_j$, one has

$$\frac{1}{H_i H_j} \partial_i \beta_{ij} = e^{-2u_i} \left[\partial_i^2 u_j + (\partial_i u_j)^2 - (\partial_i u_i)(\partial_i u_j) \right]$$

and

$$\frac{\beta_{\ell i} \beta_{\ell j}}{H_i H_j} = e^{-2u_\ell} (\partial_\ell u_i)(\partial_\ell u_j).$$

Grouping Lemma 2.4 by differentiation direction gives (4). $\square$

Equation (4) is the engine of the local theory: curvature is a sum of n channels, and the j -th channel knows only the j -th metric component in the denominator. That is the geometric precursor of the later period-theoretic split into complementary channels $\omega_\pm$.

2.3 Exact divisor-channel decomposition

Writing $u_i = \frac{1}{2} \log h_i + \frac{1}{2} \sum_a p_{ia} \log z_a$ and expanding the directional operators produces the fundamental local identity.

Theorem 2.6 (General local decomposition). *For every diagonal divisor-monomial metric germ,*

$$R[g] = \sum_{a=1}^k z^{-P_a-2e_a} F_a(z, y) + \sum_{\mu=k+1}^n z^{-P_\mu} F_\mu(z, y), \quad (5)$$

where $F_a = -2h_a^{-1}z_a^2 Q_a(u)$ and $F_\mu = -2h_\mu^{-1}Q_\mu(u)$. Every F_i is smooth up to the corner and depends only on P , the units h_i , and the first and second derivatives of those units in the single direction t^i .

Proof. Write $u_i = \frac{1}{2} \log h_i + \frac{1}{2} \sum_{a=1}^k p_{ia} \log z_a$. For a normal direction z_a ,

$$\partial_a u_i = \frac{p_{ia}}{2z_a} + \frac{1}{2} \partial_a \log h_i, \quad \partial_a^2 u_i = -\frac{p_{ia}}{2z_a^2} + \frac{1}{2} \partial_a^2 \log h_i.$$

Hence Q_a has at worst a z_a^{-2} pole and $z_a^2 Q_a$ is smooth. Since $g_a^{-1} = h_a^{-1} z^{-P_a}$,

$$-2g_a^{-1} Q_a = z^{-P_a-2e_a} F_a$$

with $F_a = -2h_a^{-1}z_a^2 Q_a(u)$. For a tangential direction $\mu > k$, the monomial factors are constant under ∂_μ , so Q_μ is smooth and

$$-2g_\mu^{-1} Q_\mu = z^{-P_\mu} F_\mu$$

with $F_\mu = -2h_\mu^{-1}Q_\mu(u)$. Summing the channels proves (5). Smoothness of each F_i up to the corner is immediate from the displayed formulae, and each F_i depends only on P , the units, and their first and second derivatives in the single direction t^i . $\square$

Corollary 2.7 (Finite base support). *The outer curvature exponents lie in the finite set $V_a = -P_a - 2e_a$ ($a \leq k$) and $V_\mu = -P_\mu$ ($\mu > k$). Dependence of the units on the normal variables produces only inward shifts $V_i + \beta$, $\beta \in \mathbb{N}^k$.*

Corollary 2.8 (Finite-jet principle). *If the first nonzero curvature coefficient occurs at inward order $|\beta| = N$, it is determined by P and a finite $(N+2)$ -jet of the units. There is no uniform jet bound after arbitrary cancellations; analytic or polyhomogeneous units give a genuine Newton expansion.*

Thus the global scalar curvature never needs to be assembled. One computes finitely many channel coefficients and reads the pole order from their exponents.

2.4 The master quadric

Fix a normal direction $a \leq k$ and write $\lambda_i = p_{ia}$. Define

$$C_a(P) = -\frac{1}{2} \left[\sum_{i \neq a} \lambda_i^2 + \sum_{\substack{r \leq s \\ r, s \neq a}} \lambda_r \lambda_s - (\lambda_a + 2) \sum_{i \neq a} \lambda_i \right]. \quad (6)$$

Theorem 2.9 (Normal-vertex coefficient).

$$F_a|_D = \frac{C_a(P)}{h_a|_D}.$$

Proof. Use

$$\partial_a u_i = \frac{\lambda_i}{2z_a} + O(1), \quad \partial_a^2 u_i = -\frac{\lambda_i}{2z_a^2} + O(1)$$

in (3). The boundary value of $z_a^2 \mathcal{Q}_a$ is

$$\frac{1}{4} \left[\sum_{i \neq a} \lambda_i^2 + \sum_{\substack{r < s \\ r, s \neq a}} \lambda_r \lambda_s - (\lambda_a + 2) \sum_{i \neq a} \lambda_i \right].$$

Multiplication by $-2/h_a$ gives $F_a|_D = C_a(P)/h_a|_D$. □

On a single face with normal exponent p_0 and transverse exponents $p_1, \dots, p_m$,

$$C(p) = -\frac{1}{2} \left[\sum_{\alpha} p_{\alpha}^2 + \sum_{\alpha < \beta} p_{\alpha} p_{\beta} - (p_0 + 2) \sum_{\alpha} p_{\alpha} \right]. \quad (7)$$

The same quadratic form supplies monomial face coefficients, corner vertices, and the structural cancellation equation $C = 0$.

2.5 Face laws

Theorem 2.10 (Single-face valuation). *On a face $x = 0$,*

$$R = x^{-(p_0+2)} F_0(x, y) + \sum_{\alpha=1}^m x^{-p_{\alpha}} F_{\alpha}(x, y),$$

with $F_0(0, y) = C(p)/h_0(0, y)$. The curvature order is the largest noncancelled member of $\{p_0 + 2 - \text{ord}_x F_0\} \cup \{p_{\alpha} - \text{ord}_x F_{\alpha}\}$.

Theorem 2.11 (Generic quadratic collapse). *If $g_x = Ax^2 + O(x^3)$ and $g_{y_{\alpha}} = P_{\alpha 0} + P_{\alpha 1}x + O(x^2)$ with $A, P_{\alpha 0} > 0$, then*

$$R = \frac{1}{A} \sum_{\alpha=1}^m \frac{P_{\alpha 1}}{P_{\alpha 0}} x^{-3} + O(x^{-2}). \quad (8)$$

The coefficient is independent of the cubic jet of g_x and of all quadratic transverse data.

Proof. Here $p_0 = 2$, every $p_{\alpha} = 0$, and $C(p) = 0$. The nominal order-four normal vertex therefore cancels. Write

$$u_0 = \log x + \frac{1}{2} \log A + O(x), \quad u_{\alpha} = \frac{1}{2} \log P_{\alpha 0} + \frac{1}{2} \frac{P_{\alpha 1}}{P_{\alpha 0}} x + O(x^2).$$

In $\mathcal{Q}_0$, the only x^{-1} term is

$$-(\partial_x u_0)(\partial_x u_{\alpha}) = -\frac{1}{2x} \frac{P_{\alpha 1}}{P_{\alpha 0}} + O(1).$$

Therefore

$$\mathcal{Q}_0 = -\frac{1}{2x} \sum_{\alpha} \frac{P_{\alpha 1}}{P_{\alpha 0}} + O(1).$$

Since $g_x^{-1} = A^{-1}x^{-2} + O(x^{-1})$, the normal channel gives (8). Tangential channels are $O(1)$, and the remaining normal terms are $O(x^{-2})$. Independence of the cubic jet of g_x and of quadratic transverse data is visible in the displayed expansions. □

Write $\mathcal{D} := \sum_{\alpha} P_{\alpha 1}/P_{\alpha 0}$ for the local transverse first-drift sum. If $\mathcal{D} = 0$, the order-three coefficient vanishes and the x^{-2} layer must be inspected; a cancelled leading term is not a pole of lower order by default.

Theorem 2.12 (Parity-fixed inverse-channel face). *If every component is even in x and $g_x = Bx^2 + O(x^4)$, $g_{y_{\alpha}} = A_{\alpha}x^{-2} + O(1)$, then*

$$R = -\frac{m(m+5)}{B}x^{-4} + O(x^{-2}). \quad (9)$$

The transverse amplitudes A_{α} do not enter the leading coefficient.

Proof. The exponent vector is $(p_0, p_1, \dots, p_m) = (2, -2, \dots, -2)$. Equation (7) gives $C(2, -2, \dots, -2) = -m(m+5)$. The normal vertex is x^{-4} ; every transverse base vertex is x^2 , hence nonpolar. Theorem 2.9 therefore supplies the leading term $-m(m+5)x^{-4}/B$. Evenness removes the inward shift by one, so the next possible normal term is x^{-2} . The transverse amplitudes A_{α} enter only the units of the transverse components and cancel from the leading normal coefficient. $\square$

Theorem 2.13 (Pure parity model). *For $d\ell^2 = Bx^2(dx)^2 + x^{-2}\sum_{\alpha} A_{\alpha}(dy_{\alpha})^2$,*

$$R = -\frac{m(m+5)}{B}x^{-4}$$

exactly.

Proof. The claim is the special case of Theorem 2.12 in which all higher jets vanish, so the $O(x^{-2})$ remainder is identically zero. For an independent check, set $r = \sqrt{B}x^2/2$. Then

$$d\ell^2 = (dr)^2 + w(r)^2 h_{\text{flat}}, \quad w \propto r^{-1/2}.$$

For an m -dimensional flat fibre,

$$R = -2m \frac{w''}{w} - m(m-1) \left(\frac{w'}{w} \right)^2.$$

Using $w'/w = -1/(2r)$, $w''/w = 3/(4r^2)$, and $4r^2 = Bx^4$ recovers the same coefficient. $\square$

Proposition 2.14 (Scalar-flat monomial locus). *For $d\ell^2 = (dr)^2 + r^{2a}\sum_{\alpha}(dy_{\alpha})^2$, $R = -m((m+1)a^2 - 2a)r^{-2}$. Thus $a = 2/(m+1)$ is exactly scalar-flat—a structural zero of the master quadric.*

2.6 Robustness and intrinsic rates

Theorem 2.15 (Divisor order is invariant). *If $x = u(x', y')x'$ with $u|_D > 0$ defines the same smooth divisor, then $\text{ord}_{x=0}(f) = \text{ord}_{x'=0}(f)$ for every scalar germ f .*

Proof. If $f = x^{\lambda}v$ with $v|_D \neq 0$, then $f = (x')^{\lambda}u^{\lambda}v$ and $u^{\lambda}v$ is again a nonzero unit on the divisor. $\square$

Theorem 2.16 (Normal-crossing robustness). *Suppose a boundary-adapted diffeomorphism preserves the components of a simple normal crossing up to permutation:*

$$z_a = u_a(z', y')z'_{\sigma(a)}, \quad u_a|_D > 0.$$

Then the multivaluation and Newton support transform only by σ ; unit factors change front coefficients but not exponent vectors. A monomial blow-up transforms exponent vectors by its integer linear map.

The diagonal representation need not remain diagonal under this change. The valuation survives because $R[g]$ is a scalar. Formula (4) may be used in any one adapted orthogonal chart without claiming that its individual directional summands are coordinate-invariant.

Proposition 2.17 (Parity change to second order). *Under $x = c(y)x' + k(y)(x')^2 + O((x')^3)$ with $c > 0$, the pure parity model has leading normal amplitude $B' = Bc^4$ and*

$$R = -\frac{m(m+5)}{Bc^4}(x')^{-4} + \frac{4m(m+5)k}{Bc^5}(x')^{-3} + O((x')^{-2}). \quad (10)$$

The order-four law is form-invariant; the displayed odd coefficient vanishes iff $k = 0$. Full parity preservation requires an odd normal coordinate change, not merely the absence of a quadratic term to one order.

Proof. The metric transformation follows from $Bx^2(dx)^2$ together with $dx = c dx' + O(x')$. Also

$$x^{-4} = c^{-4}(x')^{-4} - 4kc^{-5}(x')^{-3} + O((x')^{-2}).$$

Substitute the pure parity formula of Theorem 2.13. □

Theorem 2.18 (Proper-distance gauges). *Let d be proper normal distance to the face. For the parity-fixed face, $d = (\sqrt{B}/2)x^2 + O(x^4)$, so Theorem 2.12 gives*

$$\boxed{\lim_{x \rightarrow 0} R d^2 = -\frac{m(m+5)}{4}}. \quad (11)$$

For the generic face with drift $\mathcal{D} = \sum_{\alpha} P_{\alpha 1}/P_{\alpha 0}$, one has $d = (\sqrt{A}/2)x^2 + O(x^3)$, and Theorem 2.11 gives

$$\boxed{\lim_{x \rightarrow 0} R d^{3/2} = \frac{\mathcal{D} A^{-1/4}}{2^{3/2}}}. \quad (12)$$

These limits are the intrinsic singularity rates that survive coordinate redefinition.

Proof. For the parity-fixed face, $d \sim (\sqrt{B}/2)x^2$ implies $d^2 \sim (B/4)x^4$, so $R d^2 \sim (-m(m+5)/B x^{-4})(B/4)x^4$ tends to $-m(m+5)/4$. For the generic face, $d^{3/2} \sim (\sqrt{A}/2)^{3/2}x^3 = A^{3/4}2^{-3/2}x^3$, and multiplication by the leading term $(\mathcal{D}/A)x^{-3}$ of R yields (12). □

2.7 Newton polyhedra, corners, and cancellation

Expand $F_i = \sum_{\beta} f_{i,\beta}(y)z^{\beta}$, combine equal exponent vectors into coefficients $c_{\alpha}(y)$, and form the support $\mathcal{S}_R(y_0) = \{\alpha : c_{\alpha}(y_0) \neq 0\}$ with Newton polyhedron $\mathcal{N}_R = \text{conv}(\mathcal{S}_R + \mathbb{R}_{\geq 0}^k)$.

Theorem 2.19 (Weighted curvature valuation). *Assume analytic or polyhomogeneous coefficients, or stop at the first finite nonzero smooth jet. Along $z_a = \rho_a \epsilon^{w_a}$ with $\rho_a, w_a > 0$, write*

$$R(\rho \epsilon^w, y_0) = \sum_{\lambda} \epsilon^{\lambda} \mathcal{F}_{w,\lambda}(\rho, y_0),$$

where

$$\mathcal{F}_{w,\lambda}(\rho, y_0) = \sum_{\substack{\alpha \in \mathcal{S}_R \\ \langle w, \alpha \rangle = \lambda}} c_{\alpha}(y_0) \rho^{\alpha}.$$

Then

$$\boxed{\kappa_w(R) = -\min\{\lambda : \mathcal{F}_{w,\lambda}(\rho, y_0) \neq 0\}}. \quad (13)$$

For generic ρ this is the negative support function of $\mathcal{N}_R$. If the leading front coefficient vanishes at a special ratio, the order drops to the next nonzero face.

Proof. Substitute the monomial path, group equal powers of ϵ , and select the first nonzero group. $\square$

Theorem 2.20 (Three-dimensional corner formula). *For $g_s = h_0(s)x^{p_0}y^{q_0}$, $g_x = h_1(s)x^{p_1}y^{q_1}$, $g_y = h_2(s)x^{p_2}y^{q_2}$,*

$$R = \frac{A}{2h_1}x^{-(p_1+2)}y^{-q_1} + \frac{B}{2h_2}x^{-p_2}y^{-(q_2+2)} + L_s x^{-p_0}y^{-q_0}, \quad (14)$$

where

$$A = -p_0^2 + p_0p_1 - p_0p_2 + 2p_0 + p_1p_2 - p_2^2 + 2p_2, \quad (15)$$

$$B = -q_0^2 + q_0q_2 - q_0q_1 + 2q_0 + q_2q_1 - q_1^2 + 2q_1, \quad (16)$$

and if $a_i = \sqrt{h_i}$ with $\ell_i = a'_i/a_i$,

$$L_s = -\frac{2}{h_0} \left[\frac{a_1''}{a_1} + \frac{a_2''}{a_2} - \ell_0(\ell_1 + \ell_2) + \ell_1\ell_2 \right]. \quad (17)$$

The three candidate vertices are $V_x = (-(p_1 + 2), -q_1)$, $V_y = (-p_2, -(q_2 + 2))$, and $V_s = (-p_0, -q_0)$. Equations (15)–(16) are twice the corresponding master quadrics; L_s is the tangential channel.

Theorem 2.21 (Cancellation hierarchy). *The leading curvature order changes in exactly four ways: structural vanishing of a master or tangential coefficient; cancellation after combining equal exponent vectors; vanishing of a multi-vertex front polynomial at a special approach ratio; or higher-jet cancellation after an outer coefficient vanishes. These cases are exhaustive because (5) is a finite sum.*

Theorem 2.22 (Weighted initial forms). *Let every g_i be polyhomogeneous and fix a positive weight w . Set $\gamma_i = \text{ord}_w(g_i)$, $G_i = \text{in}_w(g_i)$, and assume the front form G_i is nonzero at the positive approach ratio under consideration. Extend w by zero on tangential coordinates and write $\tilde{w}_j$ for the resulting coordinate weights. The j -th curvature channel obeys*

$$\text{ord}_w(R_j) \geq -\gamma_j - 2\tilde{w}_j. \quad (18)$$

At the smallest candidate weight, the curvature front form is obtained by applying (3)–(4) to the weighted initial metric and retaining terms of that weight. If the sum is nonzero, it is $\text{in}_w(R)$. If it vanishes, the next filtered layer must be evaluated. Competing monomials in a single component are front-form data, not data of the exponent matrix alone.

Proof. In the w -filtration, multiplication adds orders, inversion negates order when the initial form is nonzero, and ∂_j lowers order by at most $\tilde{w}_j$. Every term of $\mathcal{Q}_j$ contains either one second derivative or two first derivatives in direction j , so $\text{ord}_w(\mathcal{Q}_j) \geq -2\tilde{w}_j$. Multiplication by g_j^{-1} gives (18). Passing the rational differential expression to the associated graded object gives the front-form rule, unless the leading sum cancels. $\square$

Example 2.23 (Two-face corner). For

$$d\ell^2 = x^{-2}y^{-2}(ds)^2 + x^2(dx)^2 + y^2(dy)^2,$$

the exponent rows are $P_s = (-2, -2)$, $P_x = (2, 0)$, $P_y = (0, 2)$. The normal master coefficients are $C_x = C_y = -6$, and the tangential coefficient vanishes because all units are constant. Thus $R = -6x^{-4} - 6y^{-4}$.

For an independent check, put $r = x^2/2$, $t = y^2/2$. Then

$$d\ell^2 = (dr)^2 + (dt)^2 + w(r, t)^2(ds)^2, \quad w = (4rt)^{-1/2}.$$

For a one-dimensional warped fibre over a flat two-dimensional base,

$$R = -2w^{-1}\Delta w = -\frac{3}{2r^2} - \frac{3}{2t^2} = -6x^{-4} - 6y^{-4}.$$

Along $x = \rho\epsilon^a$, $y = \epsilon^b$, one has $\kappa_{(a,b)}(R) = \max(4a, 4b)$. On the balanced face $a = b$, the front coefficient is $-6(\rho^{-4} + 1)$, which never cancels for $\rho > 0$.

2.8 Kerr–Newman faces and the passage to periods

Specialising the face laws recovers the local stratification used later as geometric motivation for complementary channels.

Divisor	Local type	Curvature
$T = 0$	generic quadratic collapse	order 3, coeff. from Thm. 2.11
$\Omega = 0$ or $J = 0$	parity-fixed, $m = 2$	order 4, coeff. $-14/B_J$
$\Phi = 0$ or $Q = 0$	parity-fixed, $m = 2$	order 4, coeff. $-14/B_Q$

At the double-reflection corner with exponents $P_s = (-6, 0)$, $P_x = (2, -2)$, $P_y = (-2, 2)$, the normal vertices are $V_x = (-4, 2)$, $V_y = (2, -4)$ with raw master coefficients -84 and -12 . Neither vertex cancels; along $x = \rho\epsilon^a$, $y = \epsilon$ the order is $\max(4a - 2, 4 - 2a)$, minimised at the balanced weight $a = 1$.

The local theory therefore supplies three things the rest of the paper will use: a rigorous notion of directional channel; a master coefficient that detects structural cancellation; and a concrete classification of the thermodynamic faces. What it does not supply is a comparison of complementary complete integrals built from those channels. That is the period problem.

3 Complementary channels and Landen–Trace reduction

The second scale begins where the first leaves off. Once scalar curvature has been decomposed into directional channels, one may integrate complementary channel densities along complete intervals. The resulting complete integrals are not independent: they are related by explicit degree-two isogenies to a common elliptic curve, and each differential splits into a descended trace piece and an elementary anti-translation piece. This section records that reduction as an ordinary theorem sequence inside the paper—not as a pasted research log.

3.1 Complementary complete integrals

Fix the DBP parameter field generated by

$$t = 2^{1/4} > 0, \quad s = t^2 = \sqrt{2}, \quad \epsilon \in \{+1, -1\},$$

with primary channel $\epsilon = +1$ and dual channel $\epsilon = -1$. Set

$$m_\epsilon = \frac{2 - \epsilon s}{4}, \quad n_\epsilon = \frac{4 - 3\epsilon s}{8}, \quad A_\epsilon = 3 + 2\epsilon s, \quad B_\epsilon = 2 + 2\epsilon s. \quad (19)$$

These satisfy the involution relations $m_+ + m_- = 1$, $m_+ m_- = 1/8$, $n_+ + n_- = 1$, and

$$n_\epsilon = -\frac{m_\epsilon^2}{1 - 2m_\epsilon}, \quad (20)$$

so that $n_+ < 0$ and $n_- > 1$.

Let $K(m)$ and $\Pi(n; m)$ be the standard complete elliptic integrals of the first and third kinds on $[0, 1]$. The complementary complete-integral values are

$$I_\epsilon = -t^7 [A_\epsilon \Pi(n_\epsilon; m_\epsilon) - B_\epsilon K(m_\epsilon)], \quad (21)$$

with I_- understood as a Cauchy principal value on the real path.

On the Legendre curve

$$C_\epsilon : \quad y^2 = x(x-1)(x-m_\epsilon), \quad (22)$$

the substitution $x = m_\epsilon r^2$ realises the complete interval as the upper real path $\gamma_\epsilon : x : 0 \rightarrow m_\epsilon$. Defining

$$F_\epsilon(x) = A_\epsilon \frac{m_\epsilon}{m_\epsilon - n_\epsilon x} - B_\epsilon, \quad \omega_\epsilon = -\frac{t^7}{2} F_\epsilon(x) \frac{dx}{y}, \quad (23)$$

one has $\int_{\gamma_\epsilon} \omega_\epsilon = I_\epsilon$. The pair (ω_+, ω_-) is the period-theoretic lift of the complementary channel idea from Section 2.

3.2 Degree-two isogenies onto E_{128}

Shift to the Weierstrass chart $u = x - m_\epsilon$, so

$$a_\epsilon = 2m_\epsilon - 1 = -\frac{\epsilon s}{2}, \quad b = m_\epsilon(m_\epsilon - 1) = -\frac{1}{8},$$

and C_ϵ becomes $y^2 = u^3 + a_\epsilon u^2 + bu$. Translation by the two-torsion point $T_\epsilon = (0, 0)$ is $T_\epsilon(u, y) = (b/u, -(b/u^2)y)$. The classical Vélú formulae

$$z = u + a_\epsilon + \frac{b}{u}, \quad v = y \left(1 - \frac{b}{u^2}\right) \quad (24)$$

yield $v^2 = z^3 + \epsilon s z^2 + z$. Choosing square roots $\iota_+ = 1$, $\iota_- = i$ (so $\iota_\epsilon^2 = \epsilon$) and setting

$$X = 1 + \epsilon s z, \quad Y = \iota_\epsilon t^3 v, \quad (25)$$

one obtains the common target

$$E_{128} : \quad Y^2 = X^3 - X^2 + X - 1. \quad (26)$$

Theorem 3.1 (Complementary isogenies). *The composite maps*

$$\phi_\epsilon(u, y) = \left(1 + \epsilon s \left(u + a_\epsilon + \frac{b}{u} \right), \iota_\epsilon t^3 y \left(1 - \frac{b}{u^2} \right) \right) \quad (27)$$

are degree-two isogenies $C_\epsilon \rightarrow E_{128}$. They satisfy $j(C_\epsilon) = 10976$, $j(E_{128}) = 128$, and $\Phi_2(128, 10976) = 0$.

Proof. Start from $C_\epsilon : y^2 = u^3 + a_\epsilon u^2 + bu$ with $a_\epsilon = 2m_\epsilon - 1 = -\epsilon s/2$ and $b = m_\epsilon(m_\epsilon - 1) = -1/8$. The point $T_\epsilon = (0, 0)$ has order two, and translation by it is

$$T_\epsilon(u, y) = \left(\frac{b}{u}, -\frac{b}{u^2}y \right).$$

The Vélú formulae $z = u + a_\epsilon + b/u$ and $v = y(1 - b/u^2)$ are the standard quotient by $\langle T_\epsilon \rangle$. Direct expansion of the curve equation gives

$$v^2 = z^3 + \epsilon s z^2 + z.$$

The linear change $X = 1 + \epsilon s z$, $Y = \iota_\epsilon t^3 v$ with $\iota_\epsilon^2 = \epsilon$ then maps the quotient to $Y^2 = X^3 - X^2 + X - 1$, i.e. to E_{128} . The composite is ϕ_ϵ , of degree two. The j -invariants $j(C_\epsilon) = 10976$ and $j(E_{128}) = 128$ are classical evaluations of the Weierstrass j -map on these models; the modular equation $\Phi_2(128, 10976) = 0$ is the degree-two modular polynomial relation forced by the existence of a cyclic isogeny of degree two, and is checked by direct substitution (Appendix A). $\square$

Thus the two complementary channels, defined on different curves with different moduli, become comparable on a single elliptic curve.

3.3 Trace and anti-translation

On E_{128} define the meromorphic differential

$$\Theta = 8 \frac{X - 3}{X + 7} \frac{dX}{Y}. \quad (28)$$

(This is the only use of the symbol Θ in the paper.) Write $\alpha_\epsilon := \omega_\epsilon - T_\epsilon^* \omega_\epsilon$ for the anti-invariant combination.

Theorem 3.2 (Common trace).

$$\omega_\epsilon + T_\epsilon^* \omega_\epsilon = \phi_\epsilon^* (\iota_\epsilon^{-1} \Theta), \quad (29)$$

equivalently $(\phi_\epsilon)_* \omega_\epsilon = \iota_\epsilon^{-1} \Theta$. In particular $(\phi_+)_* \omega_+ = \Theta$ and $(\phi_-)_* \omega_- = -i\Theta$.

Proof. From (24) one has $dz/v = du/y$. From (25),

$$\frac{dX}{Y} = \frac{\iota_\epsilon}{t} \frac{du}{y}. \quad (30)$$

Put $d_\epsilon = m_\epsilon(1 - n_\epsilon) = (1 + \epsilon s)/16$. In the u -coordinate,

$$F_\epsilon(u) = A_\epsilon \frac{m_\epsilon}{d_\epsilon - n_\epsilon u} - B_\epsilon. \quad (31)$$

The exact rational trace identity is

$$\boxed{F_\epsilon(u) + F_\epsilon(b/u) = -4 + \frac{40}{X+7} = -4\frac{X-3}{X+7}.} \quad (32)$$

(This is a rational identity in u , checked by clearing denominators and using $X = 1 + \epsilon s(u + a_\epsilon + b/u)$; see Appendix A.) Translation by T_ϵ preserves du/y . Therefore

$$\omega_\epsilon + T_\epsilon^* \omega_\epsilon = -\frac{t^7}{2} (F_\epsilon(u) + F_\epsilon(b/u)) \frac{du}{y} = -\frac{t^7}{2} \left(-4\frac{X-3}{X+7} \right) \frac{du}{y}.$$

Using (30) and $t^8 = 4$,

$$-\frac{t^7}{2} \cdot \left(-4\frac{X-3}{X+7} \right) \frac{du}{y} = \iota_\epsilon^{-1} \cdot 8\frac{X-3}{X+7} \frac{dX}{Y} = \phi_\epsilon^*(\iota_\epsilon^{-1}\Theta),$$

which is (29). The pushforward form $(\phi_\epsilon)_* \omega_\epsilon = \iota_\epsilon^{-1}\Theta$ is equivalent, and specialising $\epsilon = \pm 1$ gives $(\phi_+)_* \omega_+ = \Theta$ and $(\phi_-)_* \omega_- = -i\Theta$. $\square$

Before closing the anti-translation term logarithmically, record its complete-interval reduction. With $\bar{m}_\epsilon = 1 - m_\epsilon$ and $c_\epsilon = m_\epsilon/(m_\epsilon - n_\epsilon)$, the DBP parameters satisfy $d_\epsilon/(m_\epsilon - n_\epsilon) = \bar{m}_\epsilon$. Under $u = m_\epsilon(r^2 - 1)$ the translated coefficient remains on the complete interval $r \in [0, 1]$:

$$\begin{aligned} I_\epsilon - I_\epsilon^T = -t^7 A_\epsilon \left[\Pi(n_\epsilon; m_\epsilon) - \frac{c_\epsilon}{\bar{m}_\epsilon} K(m_\epsilon) \right. \\ \left. + \frac{c_\epsilon m_\epsilon}{\bar{m}_\epsilon} \Pi(\bar{m}_\epsilon; m_\epsilon) \right]. \end{aligned} \quad (33)$$

There is no incomplete amplitude: (33) is already a complete period reduction. The deeper statement is that the same anti-difference is elementary as a differential form.

Theorem 3.3 (Logarithmic anti-translation). *The anti-invariant combination satisfies $T_\epsilon^* \alpha_\epsilon = -\alpha_\epsilon$ and $(\phi_\epsilon)_* \alpha_\epsilon = 0$. Moreover, with*

$$f_\epsilon = \frac{y - \ell_\epsilon u}{y + \ell_\epsilon u}, \quad \ell_\epsilon = -\frac{2it}{\iota_\epsilon}, \quad R_\epsilon = \frac{4i}{\iota_\epsilon}, \quad (34)$$

one has the exact closure

$$\boxed{\alpha_\epsilon = R_\epsilon d \log f_\epsilon.} \quad (35)$$

In particular $\ell_+ = -2it$, $R_+ = 4i$, $\ell_- = -2t$, $R_- = 4$.

Proof. Put $u_{0,\epsilon} = d_\epsilon/n_\epsilon$ and $u_{1,\epsilon} = b/u_{0,\epsilon}$. The coefficient identity

$$F_\epsilon(u) - F_\epsilon(b/u) = -\frac{A_\epsilon m_\epsilon}{d_\epsilon} \frac{u^2 - b}{(u - u_{0,\epsilon})(u - u_{1,\epsilon})} \quad (36)$$

is elementary. Writing $G_\epsilon(u) = u^3 + a_\epsilon u^2 + bu = y^2$, the curve identities

$$uG'_\epsilon(u) - 2G_\epsilon(u) = u(u^2 - b), \quad G_\epsilon(u) - \ell_\epsilon^2 u^2 = u(u - u_{0,\epsilon})(u - u_{1,\epsilon})$$

hold once $\ell_\epsilon^2 = z_{0,\epsilon} = -4\epsilon s$. Differentiating $\log f_\epsilon$ therefore produces

$$d \log f_\epsilon = \ell_\epsilon \frac{u^2 - b}{y(u - u_{0,\epsilon})(u - u_{1,\epsilon})} du.$$

Matching the normalisation $R_\epsilon \ell_\epsilon = t^7 A_\epsilon m_\epsilon / (2d_\epsilon)$ against (23) and (36) yields (35). $\square$

Combining Theorems 3.2 and 3.3 gives the structural decomposition that organises the period layer:

$$\boxed{\omega_\epsilon = \frac{1}{2}\phi_\epsilon^*(\iota_\epsilon^{-1}\Theta) + \frac{1}{2}R_\epsilon d\log f_\epsilon.} \quad (37)$$

The first summand is the descended, non-elementary channel; the second is elementary. This is the same even/odd logic that Section 4 will execute on the entropy cover, now realised on differentials rather than on field extensions.

Because α_ϵ is principal (its residue divisor is the divisor of f_ϵ), non-torsion of the descended pole of Θ does not obstruct reduction of the anti-translation term: the two sectors are different.

3.4 Exact anti-periods and completed formulae

Along the upper complete-integral path one has the endpoint values $f_\epsilon(-m_\epsilon, 0) = -1$ and $\lim_{u \rightarrow 0^-} f_\epsilon = 1$.

Theorem 3.4 (Primary anti-period). *In the primary channel,*

$$f_+ = \frac{y + 2itu}{y - 2itu}.$$

For $-m_+ < u < 0$ on the upper oval, f_+ lies on the unit circle and its continuous argument increases from $-\pi$ to 0. Therefore $\Delta_{\gamma_+} \log f_+ = i\pi$. Using $R_+ = 4i$,

$$\boxed{I_{\text{primary}} - I_{\text{primary}}^T = -4\pi,} \quad (38)$$

equivalently $I_{\text{primary}}^T = I_{\text{primary}} + 4\pi$.

Proof. On the upper oval with $-m_+ < u < 0$, one has $y \in i\mathbb{R}_{>0}$ after the standard orientation of the complete-integral path, so $y/(itu)$ is positive imaginary over negative real and f_+ has modulus one. The continuous argument is fixed by the endpoint values $f_+(-m_+, 0) = -1$ and $\lim_{u \rightarrow 0^-} f_+ = 1$, hence runs from $-\pi$ to 0. Theorem 3.3 gives $\alpha_+ = R_+ d\log f_+$ with $R_+ = 4i$, so

$$\int_{\gamma_+} \alpha_+ = R_+ \Delta_{\gamma_+} \log f_+ = 4i \cdot i\pi = -4\pi.$$

But $\int_{\gamma_+} \alpha_+ = I_{\text{primary}} - I_{\text{primary}}^T$. □

Theorem 3.5 (Dual anti-period). *In the dual channel,*

$$f_- = \frac{y + 2tu}{y - 2tu}.$$

The real path crosses the simple zero corresponding to the third-kind pole. The real principal value is the endpoint difference of $\log |f_-|$:

$$\log |1| - \log |-1| = 0.$$

Hence

$$\boxed{\text{CPV}(I_{\text{dual}} - I_{\text{dual}}^T) = 0,} \quad (39)$$

equivalently $I_{\text{dual}}^{T, \text{CPV}} = I_{\text{dual}, \text{CPV}}$. The two one-sided logarithmic continuations satisfy

$$I_{\text{dual}} - I_{\text{dual}}^T \in \{-4\pi i, +4\pi i\},$$

and their difference has magnitude $8\pi i$.

Proof. With $R_- = 4$ real, the anti-period reduces to 4 times the increment of $\log f_-$. On the real principal-value path only the modulus contributes to the real part of the logarithm, and the endpoint values force that contribution to vanish. The imaginary one-sided values $\pm 4\pi i$ are the two choices of argument jump $\pm\pi$ multiplied by $R_- = 4$. $\square$

Let $\Gamma_\epsilon = \phi_\epsilon(\gamma_\epsilon)$, so Γ_+ runs from $X = 1$ to $+\infty$ and Γ_- from $X = 1$ to $-\infty$. The trace theorem converts sums of periods into integrals of Θ :

$$I_\epsilon + I_\epsilon^T = \iota_\epsilon^{-1} \int_{\Gamma_\epsilon} \Theta \quad (40)$$

(with CPV in the dual channel). Combining with the anti-periods yields the completed reductions

$$\boxed{I_{\text{primary}} = \frac{1}{2} \int_{\Gamma_+} \Theta - 2\pi, \quad I_{\text{dual,CPV}} = -\frac{i}{2} \text{CPV} \int_{\Gamma_-} \Theta.} \quad (41)$$

Theorem 3.6 (Pole transport and non-torsion). *Both complementary pole pairs map to $X = -7$ on E_{128} , with points*

$$P = (-7, 20i), \quad -P = (-7, -20i).$$

The residues are $\text{Res}_P(\Theta) = 4i$ and $\text{Res}_P(-i\Theta) = 4$. The point P is non-torsion: exact good reductions give

$$\text{ord}_{\mathbb{F}_{13}}(\bar{P}) = 4, \quad \text{ord}_{\mathbb{F}_{29}}(\bar{P}) = 6,$$

so any finite order N would have to satisfy both $N = 4 \cdot 13^a$ and $N = 6 \cdot 29^b$, which is impossible. Hence the pole divisor of Θ is non-torsion.

This non-torsion statement belongs to the trace sector. It does not imply non-reducibility of α_ϵ , because $(\phi_\epsilon)_ \alpha_\epsilon = 0$ and $\alpha_\epsilon = R_\epsilon d \log f_\epsilon$.*

Proof. The poles of ω_ϵ are the zeros of the denominator of F_ϵ , transported by ϕ_ϵ . Direct evaluation of ϕ_ϵ at those poles yields $X = -7$. The residue computation is the local Laurent coefficient of Θ at P , equivalently of $-i\Theta$ after the dual normalisation. For non-torsion: reduction modulo 13 and 29 is good for P , and the orders of the reduced points are computed by repeated addition in the finite-field group law (Appendix A). Incompatibility of the two order constraints forces P to have infinite order over $\mathbb{Q}(i)$. $\square$

3.5 What the period layer settles—and what it does not

The period reduction settles the internal structure of each complementary channel: isogeny to a common curve, descent of the even combination to Θ , elementary closure of the odd combination, and exact anti-period corrections. It does *not* assert a scalar algebraic relation

$$I_{\text{primary}} = \alpha(\sqrt{2}) I_{\text{dual,CPV}} + \text{elementary correction},$$

because (41) integrates Θ along different relative paths Γ_+ and Γ_- . Comparing those paths is a separate relative-homology problem. Non-torsion of P neither proves nor disproves every such relation.

Likewise, the derivation of ω_ϵ from a surface curvature two-form—the direct interface with Section 2—is a separate bridge theorem. What the present section needs, and supplies, is the reduction once the complete integrals are given.

All algebraic identities of this section are machine-checkable by a 75-gate exact verifier using no floating-point arithmetic; the gate inventory is recorded in Appendix A.

The period layer has now isolated a descended non-torsion form and an elementary correction. The global thermodynamic observables exhibit the same split: symmetric functions descend; individual roots do not. That is the Galois theory of the horizon cover.

4 Galois theory of the horizon cover

The third scale is arithmetic. The outer and inner horizon entropies are roots of a polynomial over the field of asymptotic charges. Everything that can be written rationally in those charges is a Galois invariant; everything that requires a choice of sheet is odd or mixed under the deck action. This is the same even/odd separation already seen for curvature channels and for $\omega_\epsilon = \text{trace} + \text{anti}$, now executed on the entropy cover itself.

4.1 Kerr–Newman: the quadratic cover

Solving (2) for S at fixed (M, J, Q) yields the monic quadratic

$$S^2 - e_1 S + e_2 = 0, \quad e_1 = 2\pi(2M^2 - Q^2), \quad e_2 = \pi^2(4J^2 + Q^4), \quad (42)$$

with roots

$$S_\pm = \pi(2M^2 - Q^2 \pm 2\sqrt{\Delta}), \quad \Delta = M^4 - M^2 Q^2 - J^2,$$

and $e_1^2 - 4e_2 = 16\pi^2\Delta$. Both coefficients are rational in the asymptotic charges; e_2 is mass-independent (the universal area product [4, 5] in entropy normalisation). The Galois group over $\mathbb{Q}(M, J, Q)$ is $\mathbb{Z}/2$, realised as the horizon swap $S_+ \leftrightarrow S_-$ —deck transformation $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$ of the double cover branched along the extremal locus $\Delta = 0$.

Three elementary facts organise the quadratic case. Transport around $\Delta = 0$ swaps the sheets while e_1, e_2 remain single-valued, and the branch is simple on the smooth extremal edge. Past the wall $\Delta < 0$ the roots become complex conjugates while the invariants cross rationally: the entropy never crosses; its symmetric functions do. On the two branches, $U_S(S_\pm) = \pm\sqrt{\Delta}/S_\pm$, so $T_\pm = \pm\sqrt{\Delta}/(2MS_\pm)$ and the inner temperature is negative in the Gibbsian convention [3].

Proposition 4.1 (Splitting–temperature identity). *Define the Galois-odd covariant $\vartheta := ST$ on either branch, so $\vartheta_\pm = \pm\sqrt{\Delta}/(2M)$. Then*

$$S_+ - S_- = 4\pi\sqrt{\Delta} = 8\pi M \vartheta_+.$$

Under the horizon swap, ϑ is odd, while temperature is mixed: $\tau(T_+) = T_-$. For $e_2 \neq 0$ the temperatures form a conjugate pair with rational symmetric functions

$$T_+ + T_- = -\frac{2\pi\Delta}{Me_2}, \quad T_+ T_- = -\frac{\Delta}{4M^2 e_2}.$$

At $J = Q = 0$ one recovers the Schwarzschild value $T_+ = 1/(8\pi M)$.

The separation of the sheets is the odd covariant. The invariant ring sees ϑ only through its square; the sign that distinguishes outer from inner is precisely what the Galois quotient forgets. Temperature inherits that arrow through $T = \vartheta/S$, which is how the negative inner temperature arises, but T itself is not of pure parity.

4.2 Left/right dictionary from the grading

Define sector entropies

$$S_L = \frac{1}{2}(S_+ + S_-) = \frac{1}{2}e_1 = \pi(2M^2 - Q^2), \quad S_R = \frac{1}{2}(S_+ - S_-) = 2\pi\sqrt{\Delta},$$

and sector temperatures by conjugacy at fixed (J, Q) : $1/T_{L,R} := \partial_M S_{L,R}$. One finds exactly

$$T_L = \frac{1}{4\pi M}, \quad T_R = \frac{\sqrt{\Delta}}{Me_1}, \quad \frac{1}{T_{\pm}} = \frac{1}{T_L} \pm \frac{1}{T_R}, \quad (43)$$

$$\left(\frac{T_R}{T_L}\right)^2 = 1 - \frac{4e_2}{e_1^2} = \frac{S_R^2}{S_L^2}. \quad (44)$$

The left/right asymmetry is the normalised discriminant. The Cardy-consistent hidden-conformal temperatures of [8] are $e_1/(8\pi^2 J)$ and $\sqrt{\Delta}/(2\pi J)$ in the present variables, with $c_L = c_R = 12J$, and the Cardy formula returns S_L and S_R . Extremal chirality [10] is the death of the odd sector at the branch point.

Proposition 4.2 (Sector laws and charge parity). *With $\Omega_X = -T_X \partial_J S_X$ and $\Phi_X = -T_X \partial_Q S_X$,*

$$\Omega_L \equiv 0, \quad \Phi_L = \frac{Q}{2M}, \quad M = 2T_L S_L + \Phi_L Q, \quad (45)$$

$$\Omega_R = \frac{J}{M(2M^2 - Q^2)}, \quad \Phi_R = \frac{MQ}{2M^2 - Q^2}, \quad M = 2T_R S_R + 2\Omega_R J + \Phi_R Q. \quad (46)$$

Angular momentum couples only to the odd sector; charge straddles both.

Observation 4.3 (Charge-parity principle). A conserved charge couples to the sectors according to which symmetric functions carry it. J appears only in e_2 (equivalently only in Δ): it is parity-pure and $\Omega_L \equiv 0$. Q leaks into e_1 : it straddles. This is why the J -picture of hidden conformal symmetry is grading-natural, while charged pictures require an auxiliary scale.

At the branch point, $S_{\text{ext}} = e_1/2 = \sqrt{e_2}$: the extremal entropy is Vieta's formula at discriminant zero—the attractor value [14] as an algebraic identity. Electromagnetic duality is the symmetry group of e_2 .

4.3 Four charges: the invariant persists, the cover jumps

The four-charge Cvetič–Youm class [11, 12] is parametrised by a seed mass m , rotation a , and boosts δ_i :

$$\begin{aligned} M &= \frac{m}{4} \sum_i \cosh 2\delta_i, & Q_i &= \frac{m}{4} \sinh 2\delta_i, & J &= ma(\Pi_c - \Pi_s), \\ N_i &= 4Q_i, & S_{\pm} &= 2\pi m[m(\Pi_c + \Pi_s) \pm \sqrt{m^2 - a^2}(\Pi_c - \Pi_s)], \end{aligned} \quad (47)$$

with $\Pi_c = \prod_i \cosh \delta_i$ and $\Pi_s = \prod_i \sinh \delta_i$. The zero-boost and equal-charge limits recover the Kerr and Kerr–Newman quadratics.

Proposition 4.4 (Four-charge invariant). *As an unconditional algebraic identity,*

$$e_2 = S_+ S_- = \pi^2(4J^2 + N_1 N_2 N_3 N_4).$$

The Kerr–Newman Q^4 is the equal-charge diagonal of the four-fold product. The static sectors $S_{L,R} = 2\pi m^2(\Pi_c \pm \Pi_s)$ realise the Galois even/odd split as the cosh/sinh product split of left/right excitation counting [1, 13].

Eliminating the boosts produces the four-radical mass relation

$$4M = \sum_{i=1}^4 \sqrt{m^2 + N_i^2}, \quad (48)$$

whose norm form

$$\mathcal{N}(u) = \prod_{\varepsilon \in \{\pm 1\}^4} \left(4M - \sum_i \varepsilon_i \sqrt{u + N_i^2} \right), \quad u = m^2, \quad (49)$$

is a polynomial of degree exactly five in u , with leading coefficient $-2^{24}M^6$ independent of the charges (Appendix B).

Proposition 4.5 (Charge ladder). *For n charges the analogous norm has u -degree $1, 1, 4, 5$ at $n = 1, 2, 3, 4$. At $n \leq 2$ the seed is rational in the observables (at two charges it is the product of the four BPS walls); at $n = 3$ one has a solvable quartic; $n = 4$ is the first quintic. The solvability threshold sits exactly at four charges.*

4.4 Abel–Ruffini for mass and entropy

Theorem 4.6 (S_5 obstruction for the seed mass). *Over $\mathbb{Q}(M, Q_1, \dots, Q_4)$ the polynomial $\mathcal{N}(u)$ is irreducible with Galois group S_5 . Consequently $u = m^2$ admits no closed-form radical expression in the physical variables, and the boost parametrisation cannot be inverted in radicals.*

Computer-certified. By the specialisation principle, the Galois group of any separable rational specialisation embeds in the generic group; hence a single specialisation with non-solvable group forces the generic group to be non-solvable, and, being a transitive subgroup of S_5 witnessing S_5 cycle types, to equal S_5 . At three independent rational points—including the integer charge vector $M = 15$, $Q = (1, 2, 3, 5)$, chosen directly rather than through the parametrisation, above the BPS bound $M > \sum_i Q_i$ —the specialised quintic is separable and irreducible over $\mathbb{Q}$, the physical (all-plus branch) root lies on it, and Dedekind reduction exhibits Frobenius elements of cycle type $(2, 3)$ at $p = 53, 23, 31$ respectively. A $(2, 3)$ element has order six and exists in no transitive quintic group except S_5 . All five roots are real at each point (the quintics are totally real): the physical seed has four real Galois-conjugate siblings over the same observables. Witnesses are recorded in Appendix A. $\square$

The obstruction propagates from mass to entropy: the radical tower above the quintic collapses.

Lemma 4.7 (Collapse of the radical tower; static case). *Let $w_i = \sqrt{m^2 + N_i^2}$ and $u = m^2$. Then, exactly:*

1. *the branch-compatible signed radicals lie in the quintic field—there exist $g_i \in \mathbb{Q}[u]/(\mathcal{N})$ with $g_i^2 \equiv u + N_i^2$ and $\sum_i g_i \equiv 4M$ (certified as polynomial identities modulo $\mathcal{N}$);*
2. *with $A = \prod_i (w_i + m)$ and $B = \prod_i (w_i - m)$ one has $AB = \prod_i N_i^2 \in \mathbb{Q}$, so that $y := (S_+ + S_-)/\pi = \sqrt{A} + \sqrt{B}$ satisfies*

$$y^2 = \gamma(u) := 2[e_4(w) + u e_2(w) + u^2 + \prod_i N_i] \in \mathbb{Q}[u]/(\mathcal{N}).$$

Theorem 4.8 (Entropy non-radical; static four-charge). *The element γ is a primitive element of the quintic field: its resolvent $\text{Res}_u(\mathcal{N}, X - \gamma)$ is an irreducible quintic whose Galois group is again S_5 (Dedekind witnesses of type $(2, 3)$ at $p = 17, 19, 23$). Since $y^2 = \gamma$, the field $\mathbb{Q}(M, \vec{Q})(S_+ + S_-)$ contains the quintic field and its Galois closure surjects onto S_5 : it is non-solvable. Hence, in the physical variables:*

1. $S_+ + S_-$ admits no radical expression; its exact degree is 10 (the polynomial $\text{Res}_u(\mathcal{N}, Y^2 - \gamma)$ is irreducible);
2. $S_+ S_- = \pi^2 \prod_i N_i$ is rational, so S_+ and S_- individually admit no radical expression, and neither do S_L and S_R ;
3. $S_L^2 = (\pi^2/4)\gamma$: the left-sector level count is itself a primitive element of the unsolvable field, while the level-matched combination $S_L^2 - S_R^2 = e_2$ is rational.

Proof. Lemma 4.7 places $y^2 = \gamma$ in the quintic field. The resolvent computation (Appendix A) shows that γ generates that field and that the resolvent is an irreducible S_5 quintic by the same Dedekind criterion used in Theorem 4.6. The three enumerated consequences are then purely field-theoretic: adjoining a square root of a primitive element of an S_5 extension yields a non-solvable extension of degree dividing 10, and irreducibility of $\text{Res}_u(\mathcal{N}, Y^2 - \gamma)$ forces the degree to be exactly 10; rationality of the product is Proposition 4.4 at $J = 0$; the identity $S_L^2 - S_R^2 = e_2$ is elementary from the definitions of the sectors. $\square$

Remark 4.9 (Methodological consequence). The left/right apparatus of [1, 2, 3] is built by manipulating $S_{\pm}$ in the boost frame. Theorems 4.6 and 4.8 show that at four charges those roots cannot be transported to the observables even in principle. The quantities expressible in the observables are precisely the Galois invariants. *Universality is descent: universal formulae are universal because they are norms.*

Proposition 4.10 (Wall factorisation). *The product of the five mass-fiber branches is*

$$\prod_{k=1}^5 u_k = \frac{256}{M^6} \prod_{\varepsilon \in \{\pm 1\}^4} (M + \varepsilon_1 Q_1 + \varepsilon_2 Q_2 + \varepsilon_3 Q_3 + \varepsilon_4 Q_4).$$

The Galois norm vanishes exactly on the sixteen marginal-stability walls, and is positive throughout the physical chamber $M > \sum_i |Q_i|$.

Thus the invariant tower is more rational than solvability requires: the roots need S_5 ; the quadratic invariant is rational and quantized; the top invariant of the quintic splits completely over $\mathbb{Q}$ into BPS walls.

4.5 Kummer modules and the degree-20 crown

Above each mass sheet lie square roots representing axial sign, entropy sum, and entropy difference. These are controlled by conjugate Kummer theory.

Theorem 4.11 (Conjugate Kummer module). *Let K/F be finite separable of degree d , let E/F be its normal closure, put $G = \text{Gal}(E/F)$, and let $I = \text{Hom}_F(K, E)$. For nonzero $r_1, \dots, r_s \in K$, write $r_{\alpha,i} = i(r_{\alpha})$ and set*

$$H = K(\sqrt{r_1}, \dots, \sqrt{r_s}), \quad L = E(\sqrt{r_{\alpha,i}} : \alpha \leq s, i \in I).$$

The conjugate Kummer module is the image W of

$$\vartheta_{\text{Kum}} : \mathbf{F}_2^s \otimes \mathbf{F}_2[I] \longrightarrow E^*/E^{*2}, \quad e_{\alpha,i} \longmapsto [r_{\alpha,i}]. \quad (50)$$

Then L is the normal closure of H/F , and restriction gives

$$1 \longrightarrow W^{\vee} \longrightarrow \text{Gal}(L/F) \longrightarrow G \longrightarrow 1.$$

If all sd conjugate square classes are independent, then

$$\text{Gal}(L/F) \cong (C_2^s)^I \rtimes G = C_2^s \wr_I G, \quad |\text{Gal}(L/F)| = 2^{sd} |G|.$$

Proof. Every F -embedding of H sends $\sqrt{r_\alpha}$ to a signed square root of some $r_{\alpha,i}$, so its normal closure contains L . Conversely the displayed generators are permuted up to sign by every F -embedding, hence L/F is normal. Kummer theory over E identifies the kernel with $W^\vee$. In the independent case the signed-permutation representation into $(C_2^s)^I \rtimes G$ is injective and the orders agree. $\square$

A discrete valuation v of E supplies the functional $[x] \mapsto v(x) \bmod 2$. Thus a parity matrix

$$A = (v_q(r_{\alpha,i}) \bmod 2) \quad (51)$$

of full column rank proves independence. If s divisor types are supported on one distinguished sheet and their parity rows form $B \in \mathrm{GL}_s(\mathbf{F}_2)$, transitivity of G transports them to every sheet and produces $B \otimes I_d$.

Return to $F = \mathbb{Q}(M, N_1, \dots, N_4)$, $K = F(u)$, $P = \prod_i N_i$, and

$$\alpha = e_4(w) + ue_2(w) + u^2, \quad \beta = e_3(w) + ue_1(w), \quad \gamma = 2(\alpha + P), \quad (52)$$

so that $\alpha^2 - u\beta^2 = P^2$ and $\gamma(\gamma - 4P) = 4u\beta^2$. For a sign vector ε , let $\ell_\varepsilon = 4M - \sum_i \varepsilon_i N_i$ and $s_\varepsilon = \prod_i \varepsilon_i$. At its signed contact component, $u = 0$ and $w_i = \varepsilon_i N_i$. Away from $\sum_i \varepsilon_i / N_i = 0$, the valuation parities are

s_ε	$v(u)$	$v(\gamma)$	$v(\gamma - 4P)$
+1	1	0	1
-1	1	1	0

These contact valuations supply the independent parity rows for $[u]$ and $[\gamma]$.

Theorem 4.12 (Degree-20 crown). *Generically $[u]$ and $[\gamma]$ are independent in K^*/K^{*2} , so $[K(\sqrt{u}, \sqrt{\gamma}) : F] = 20$. Either ordered horizon generates this field.*

Proof. An even contact valuation proves that u and $u\gamma$ are nonsquares; an odd contact valuation proves that γ is a nonsquare. If $y = \hat{S}_+ + \hat{S}_-$ and $z = \hat{S}_+ - \hat{S}_-$ with $\hat{S}_\pm = S_\pm/\pi$, then $y^2 = \gamma$ and $z = 2\beta\sqrt{u}/y$. Primitivity $K = F(\gamma)$ from Theorem 4.8, together with $\hat{S}_+\hat{S}_- = P \in F$, shows that either horizon recovers the other and then the entire field. $\square$

Transporting the contact valuations around the S_5 orbit produces the parity matrix

$$V_{\mathrm{st}} = \begin{pmatrix} I_5 & 0 \\ I_5 & I_5 \end{pmatrix} \quad (53)$$

on the columns $(u_1, \dots, u_5 \mid \gamma_1, \dots, \gamma_5)$. Full rank implies independence of the ten conjugate square classes.

Theorem 4.13 (Static wreath closure).

$$\mathrm{Gal}(\tilde{H}/F) \cong C_2^2 \mathrm{wr} S_5, \quad |\mathrm{Gal}(\tilde{H}/F)| = 2^{10} \cdot 120 = 122880.$$

The action has five mass-sheet blocks of size four; in each block, V_4 acts regularly on $\{\hat{S}_+, \hat{S}_-, -\hat{S}_+, -\hat{S}_-\}$.

Proof. Exact even and odd wall witnesses show that the selected contacts are not components of the quintic discriminant, hence their valuations are unramified in the S_5 splitting field E/F . Their S_5 -orbits produce the matrix (53), which is invertible over $\mathbf{F}_2$. Theorem 4.11 with $s = 2$ and $G = S_5$ therefore yields the wreath product of order $2^{10} \cdot 120$. The regular V_4 -action on each mass-sheet block is the standard action of the Klein four-group on the four signed ordered horizons. $\square$

4.6 Rotation, inertia, and realisation

Let J be transcendental over F and set $\delta_J = \gamma - 4P - 16J^2$. In the boost frame the entropy sum remains independent of rotation while the difference acquires J :

$$\frac{S_+ + S_-}{\pi} = 4m^2(\Pi_c + \Pi_s), \quad \frac{S_+ S_-}{\pi^2} = P + 4J^2, \quad (54)$$

so $y^2 = \gamma$ is J -independent and $z^2 = \delta_J$.

Theorem 4.14 (Rank jump under rotation). *The classes $[u]$, $[\gamma]$, and $[\delta_J]$ are independent in $K(J)^*/K(J)^{*2}$. Consequently the ordered-horizon field over $F(J)$ still has degree 20, while the fully augmented field $K(J)(\sqrt{u}, \sqrt{\gamma}, \sqrt{\delta_J})$ has degree 40. At $J = 0$ one has $[\delta_0] = [u][\gamma]$, and the third Kummer channel collapses onto the static crown.*

Proof. Pure transcendental extension preserves the quintic and the old square classes. By the crown identity, $[\gamma - 4P] = [u][\gamma]$, so $\gamma - 4P$ is a nonsquare. Hence $\delta_J = (\gamma - 4P) - 16J^2$ is irreducible in $K[J]$ and its height-one prime has odd valuation on δ_J but zero valuation on the two constant classes. At $J = 0$, $\sqrt{\delta_0} = 2\beta\sqrt{u}/\sqrt{\gamma}$. $\square$

Choosing one private prime per sheet for the rotating differences yields the parity matrices

$$V_{\text{aug}} = \begin{pmatrix} I_5 & 0 & 0 \\ I_5 & I_5 & 0 \\ 0 & 0 & I_5 \end{pmatrix}, \quad V_{\text{hor}} = \begin{pmatrix} I_5 & 0 \\ 0 & I_5 \end{pmatrix}, \quad (55)$$

both invertible.

Theorem 4.15 (Rotating wreath closures).

$$\text{Gal}(\tilde{H}_J/F_J) \cong C_2^2 \text{ wr } S_5, \quad \text{Gal}(\tilde{L}_J/F_J) \cong C_2^3 \text{ wr } S_5,$$

of orders 122880 and 3932160 respectively.

Proof. Both matrices in (55) are invertible over $\mathbf{F}_2$. Theorem 4.11 applies with base group S_5 and $s = 2$ (ordered horizon) or $s = 3$ (augmented). $\square$

Tame inertia is read from the same parity matrices. Locally write $r_{\alpha,i} = \varpi^{m_{\alpha,i}} \epsilon_{\alpha,i}$. Away from residue characteristic two, generic divisorial Kummer inertia is cyclic of order two and flips precisely the square roots with $m_{\alpha,i}$ odd. For a wreath element $(a; \sigma)$, each cycle c of σ carries colour $\kappa_c = \sum_{i \in c} a_i$; a cycle of length ℓ has order ℓ when $\kappa_c = 0$ and order 2ℓ otherwise. The generic catalogue is:

Locus	Parity/colour	Generic inertia
simple mass branch	uncoloured transposition	C_2
even signed contact	axial $(1, 0, 0)$	C_2
odd signed contact	axial-sum $(1, 1, 0)$	C_2
rotating difference $\delta_i = 0$	difference $(0, 0, 1)$	C_2
reciprocal sub-balance	coloured transposition	C_4
$J = 0$	Kummer relation divisor	unramified rank drop
$N_i^2 = N_j^2$	normalised sheet crossing	unramified

These strata are realised on the incidence cover

$$X : \quad w_i^2 - u - N_i^2 = 0 \ (1 \leq i \leq 4), \quad \sum_i w_i = 4M, \quad (56)$$

with relative Jacobian $8e_3(w)$. After saturation by $2MP \prod_{a < b} (N_a^2 - N_b^2)$, every named stratum is nonempty:

Stratum	codim in X	degree
mass ramification I_R	1	48
even/odd contact	1	2
rotating difference I_Z	1	64
$I_R + I_Z$	2	192
reciprocal sub-balance	2	6
contact+ I_Z	2	8
I_R +contact+ I_Z	3	24

Theorem 4.16 (Realisation-poset certificate). *After saturation by $2MP \prod_{a < b} (N_a^2 - N_b^2)$, every named stratum is nonempty with the displayed codimension and degree. The difference ideal*

$$I_Z = I_X + (P + 8J^2 - u^2 - ue_2(w) - e_4(w)) \quad (57)$$

is prime, reduced, and irreducible over $\mathbb{Q}$; its minimal-prime and primary decompositions each consist of I_Z alone. The intersection $I_R + I_Z$ has a single irreducible support. Even contact-difference intersections are supported on $J = 0$ with multiplicity two.

Computer-certified. The incidence ideal I_X and the difference ideal I_Z are constructed in Macaulay2 from (56) and (57). After the displayed saturation, codimension and degree computations reproduce the table above. The commands `minimalPrimes IZ` and `primaryDecomposition IZ` each return a single component equal to I_Z , of ambient codimension six and degree 64. Details are recorded in Appendix A. $\square$

Ramification must be computed on this normalised cover: the discriminant of the projected quintic acquires extraneous squared factors at sheet crossings and is not the true branch divisor.

The global arithmetic layer is now complete: invariants descend, roots do not, and the decorated cover above the S_5 base is a controlled wreath product. Section 5 records the three-scale comparison as a single picture.

5 The three-scale dictionary

The preceding sections are not three independent monographs. Each isolates a carrier, reduces by an even/invariant operation, and retains an odd/root residual. Table 1 and Figure 1 record the comparison.

Three structural analogies organise the dictionary.

Channel versus sheet. Locally, curvature is a sum of directional channels (4). At the period scale each complementary differential splits as (37) into a descended trace and an elementary anti term. Globally, the entropy pair splits into an even sector S_L and an odd sector S_R . In every case one decomposes, pushes the even part down, and retains the odd part as monodromy or path data.

Table 1: Object dictionary across the three scales.

Scale	Carrier	What descends	What remains odd/root
Local curvature	diagonal divisor-monomial germ near D	pole orders $\kappa_w(R)$, master coefficients $C_a(P)$, rates Rd^2 , $Rd^{3/2}$	individual channel summands $\mathcal{Q}_j/g_j$ (chart-dependent)
Period reduction	complementary differentials $\omega_\pm$ on $C_\pm$	pushforwards $(\phi_\epsilon)_*\omega_\epsilon = \iota_\epsilon^{-1}\Theta$	anti-translation $\alpha_\epsilon = R_\epsilon d \log f_\epsilon$; path choice $\Gamma_\pm$
Galois cover	horizon pair (S_+, S_-) over observables	e_1, e_2 , wall product, S_L , level-matched differences	$\sqrt{\Delta}$, S_R , sign of $\vartheta = ST$, individual $S_\pm$

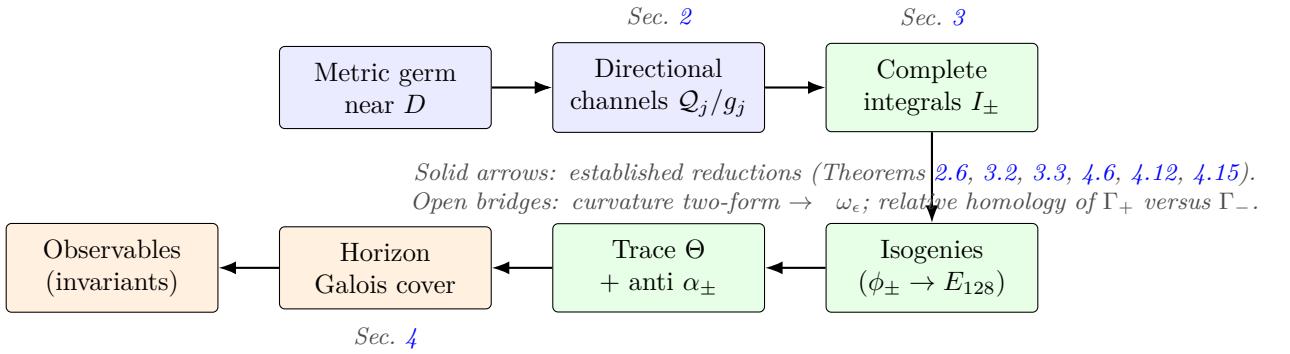


Figure 1: Sequential pipeline of the paper. Each solid arrow is a theorem of Sections 2–4; the two open bridges are stated as such in the text.

Valuation parity versus Galois parity. The master quadric $C_a(P)$ is a quadratic form on one column of the metric valuation matrix. Kummer independence is a parity matrix of divisorial valuations. Both convert a geometric configuration into a linear-algebraic rank computation.

Cancellation versus completion. Theorem 2.21 lists the four ways a local pole order can drop. Section 3 separates completed channel reductions from the open comparison of Γ_+ with Γ_- . Theorem 4.8 separates certified non-radicality from uncertified inheritance for sector temperatures. In each case the paper records exact closure status rather than silent over-claim.

The two open interface theorems are therefore not failures of the spine but its boundary:

1. deriving ω_ϵ from the original surface curvature two-form and matching poles to the local face table;
2. comparing the relative homology classes of Γ_+ and Γ_- on E_{128} .

6 Discussion and open problems

The multi-horizon story, read at three scales, is a single even/odd principle. Locally, pole orders and master coefficients descend as scalar invariants while individual curvature channels remain chart-dependent. At the period scale, the even combination of complementary differentials descends to a non-torsion form Θ on E_{128} , while the odd combination is elementary. Globally, area products, wall data, and left-sector half-energy descend to the observables, while individual entropies and the sign of $\vartheta = ST$ live only on the cover. At four charges a non-solvable wall separates the two layers: the solution-generating map is radical-one-way.

The classical checks of the hidden-conformal programme are reproduced by the Galois grading alone, with no conformal input. Those checks therefore carry no discriminating power about the existence of a dual CFT; the burden of evidence rests on genuinely quantum observables.

The decorated arithmetic cover is now closed as a degree-20 crown with full C_2^2 wr S_5 normal closure, and rotation supplies a controlled third Kummer channel. Open problems lie higher in the tower or on the two bridges already named:

1. extend contact-parity realisation and maximal wreath theorems to every charge number with known base monodromy;
2. determine the lower symmetric functions of the mass fiber;
3. certify sector-temperature fields;
4. interpret Kummer characters and inertia colours microscopically;
5. classify compactified higher-collision strata;
6. treat five-dimensional and cosmological-constant families;
7. derive ω_ϵ from the surface curvature two-form and match to Section 2;
8. compare Γ_+ and Γ_- in the relative homology of $(E_{128}, \{P, -P\})$.

A Computational certificates

All machine-checked claims are exact: rational arithmetic in the Python standard library, symbolic identities in `sympy`, or Macaulay2 primary decompositions. Numerics appear only as scaffolding and are discharged by exact polynomial identities.

Script	Layer	Contents
<code>verify_local_curvature_calculus.py</code>	local	75 rational two-jet checks of (4); face laws; pure models; KN coefficients
<code>verify_dbp_landen_trace_theorem_v1_1.py</code>	periods	75 exact gates: isogeny, trace, anti, residues, non-torsion
<code>four_charge_crown.py</code>	Galois	Prop. 4.4; anchors
<code>abel_ruffini_black_holes.py</code>	Galois	Thm. 4.6; Dedekind witnesses
<code>entropy_sum_resolvent.py</code>	Galois	Lem. 4.7; Thm. 4.8
<code>horizon_wreath_inertia_model.m2</code>	Galois	realisation poset; primality of I_Z

Dedekind witnesses (Theorem 4.6). Point A: $M = 3$, $Q = (\frac{1}{3}, \frac{1}{2}, \frac{2}{5}, \frac{3}{7})$; witness $p = 53$, type $(2, 3)$. Point B: $M = 15$, $Q = (1, 2, 3, 5)$; witness $p = 23$. Point C: $M = 2$, $Q = (\frac{2}{7}, \frac{5}{9}, \frac{1}{4}, \frac{4}{11})$; witness $p = 31$. All quintics separable and totally real.

Entropy resolvent (Theorem 4.8, point B). Exact certificates modulo $\mathcal{N}$: $g_i^2 \equiv u + N_i^2$, $\sum g_i \equiv 4M$, $\alpha^2 - u\beta^2 \equiv (\prod N_i)^2$. Resolvent witnesses $p = 17, 19, 23$, all type $(2, 3)$; degree-10 polynomial for $S_+ + S_-$ irreducible over $\mathbb{Q}$.

Macaulay2. Under Macaulay2 1.25.11: $\text{codim } I_X = 5$, relative Jacobian $8e_3(w)$, and both `minimalPrimes IZ` and `primaryDecomposition IZ` return the single prime I_Z of degree 64.

B Degree count

For large u each factor of (49) behaves as $4M - (\sum_i \varepsilon_i)\sqrt{u} + O(u^{-1/2})$. Pairing ε with $-\varepsilon$: the aligned pair contributes degree 1, the four pairs with $|\sum \varepsilon| = 2$ contribute degree 4, and the balanced pairs contribute degree 0 in u . Hence $\deg_u \mathcal{N} = 5$ and $\text{lc} = (-16)(-4)^4(4M)^6 = -2^{24}M^6$. The same count gives degrees 1, 1, 4 at $n = 1, 2, 3$ charges.

C Local computation recipe

Given diagonal components g_i , valuation matrix P , and unit/front germs: verify positivity; form channels $\mathcal{Q}_j$; apply (5); evaluate $C_a(P)/h_a$; combine equal exponents; inspect inward jets only if an outer coefficient vanishes; select the Newton face; evaluate the front coefficient at the declared ratio; report pole order and leading coefficient. Numerical curvature is a consistency check, not the source of the valuation.

D Reproducibility record

Item	Record
Document	v2.1 (13 July 2026), sequential unified preprint
Author	William Lloyd
Mathematical sources	local curvature calculus v1.0; DBP Landen–Trace v1.1; Galois horizon cover v1.0
Presentation	rewritten as one narrative (not a verbatim merge)
Notation	Θ : DBP trace on E_{128} ; $\vartheta = ST$: Galois-odd; $\mathcal{D}$: local transverse drift
Curvature convention	unit S^2 has $R = +2$
Units	$G = \hbar = c = 1$, $S = A/4$

References

- [1] M. Cvetič and F. Larsen, “General rotating black holes in string theory: Greybody factors and event horizons,” *Phys. Rev. D* **56** (1997) 4994, [hep-th/9705192].
- [2] M. Cvetič and F. Larsen, “Greybody factors for rotating black holes in four dimensions,” *Nucl. Phys. B* **506** (1997) 107, [hep-th/9706071].
- [3] M. Cvetič, G. W. Gibbons, H. Lü and C. N. Pope, “Killing Horizons: Negative Temperatures and Entropy Super-Additivity,” *Phys. Rev. D* **98** (2018) 106015, [arXiv:1806.11134].
- [4] M. Ansorg and J. Hennig, “The inner Cauchy horizon of axisymmetric and stationary black holes with surrounding matter,” *Class. Quant. Grav.* **25** (2008) 222001, [arXiv:0810.3998]; *Phys. Rev. Lett.* **102** (2009) 221102, [arXiv:0903.5405].
- [5] M. Cvetič, G. W. Gibbons and C. N. Pope, “Universal Area Product Formulae for Rotating and Charged Black Holes in Four and Higher Dimensions,” *Phys. Rev. Lett.* **106** (2011) 121301, [arXiv:1011.0008].
- [6] A. Castro and M. J. Rodriguez, “Universal properties and the first law of black hole inner mechanics,” *Phys. Rev. D* **86** (2012) 024008, [arXiv:1204.1284].
- [7] A. Castro, A. Maloney and A. Strominger, “Hidden Conformal Symmetry of the Kerr Black Hole,” *Phys. Rev. D* **82** (2010) 024008, [arXiv:1004.0996].
- [8] Y.-Q. Wang and Y.-X. Liu, “Hidden Conformal Symmetry of the Kerr-Newman Black Hole,” *JHEP* **08** (2010) 087, [arXiv:1004.4661].
- [9] K.-N. Shao and Z. Zhang, “Hidden Conformal Symmetry of Rotating Black Hole with four Charges,” *Phys. Rev. D* **83** (2011) 106008, [arXiv:1008.0585].
- [10] M. Guica, T. Hartman, W. Song and A. Strominger, “The Kerr/CFT Correspondence,” *Phys. Rev. D* **80** (2009) 124008, [arXiv:0809.4266].
- [11] M. Cvetič and D. Youm, “Entropy of non-extreme charged rotating black holes in string theory,” *Phys. Rev. D* **54** (1996) 2612, [hep-th/9603147].
- [12] Z.-W. Chong, M. Cvetič, H. Lü and C. N. Pope, “Charged rotating black holes in four-dimensional gauged and ungauged supergravities,” *Nucl. Phys. B* **717** (2005) 246, [hep-th/0411045]; cf. also arXiv:1912.08988.

- [13] A. Strominger and C. Vafa, “Microscopic origin of the Bekenstein-Hawking entropy,” *Phys. Lett. B* **379** (1996) 99, [hep-th/9601029].
- [14] G. W. Moore, “Arithmetic and Attractors,” [hep-th/9807087].
- [15] N. Benjamin, S. Kachru, K. Ono and L. Rolen, “Black holes and class groups,” *Res. Math. Sci.* (2018), [arXiv:1807.00797]; M. Günaydin, S. Kachru and A. Tripathy, [arXiv:1903.02323].
- [16] C.-M. Chen, Y.-M. Huang, J.-R. Sun, M.-F. Wu and S.-J. Zou, “Twofold Hidden Conformal Symmetries of the Kerr-Newman Black Hole,” *Phys. Rev. D* **82** (2010) 066004, [arXiv:1006.4097].
- [17] D. S. Dummit, “Solving solvable quintics,” *Math. Comp.* **57** (1991) 387.
- [18] D. R. Grayson and M. E. Stillman, “Macaulay2,” <https://macaulay2.com/>.